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Proton Belt Dynamics Following the May 2024 Geospace Magnetic Superstorm



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Introduction and Observations

In May 2024, a massive sunspot region produced a series of intense solar flares and coronal mass ejections. This resulted in an extreme **G5 geomagnetic storm**, the largest to hit Earth in 2 decades, with the disturbance storm time index (Dst) dropping below -412 nT , which coincided with a significant solar radiation storm.

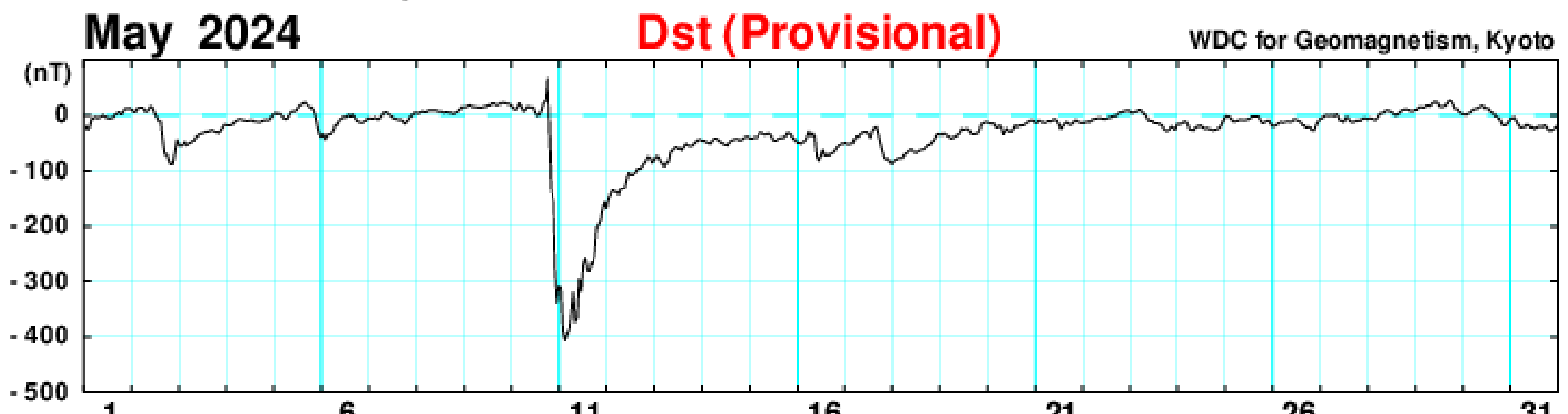


Figure 2: Provisional Dst index obtained from the Kyoto World Data Center for May 2024.

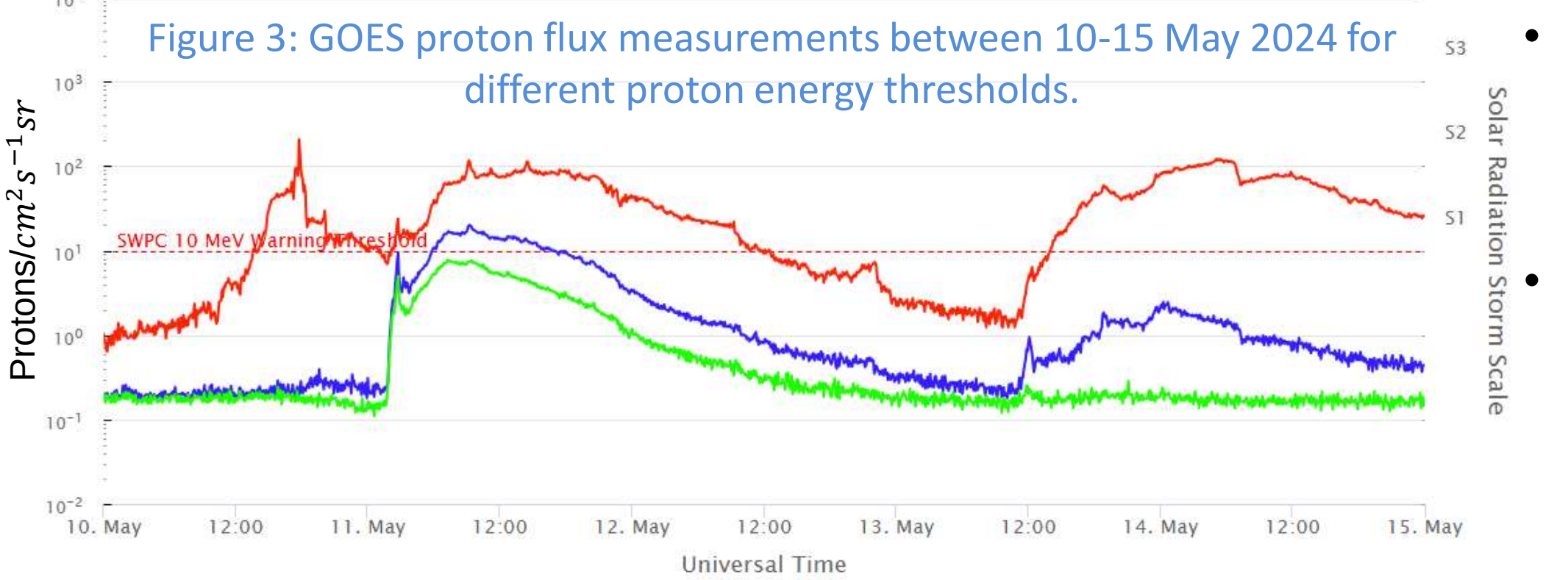


Figure 3: GOES proton flux measurements between 10-15 May 2024 for different proton energy thresholds.

- The flux of protons with energies greater than 10 MeV exceeded the standard SEP threshold.
- The event was also visible at substantially higher proton energies: Elevated fluxes were observed in the >50 MeV and >100 MeV channels.

The Mother's Day superstorm led to a pronounced enhancement of proton fluxes in the outer edge of the Earth's inner radiation belt, that was observed by several satellites.

To study this enhancement, we utilise data from two missions that carry ESA's **Next Generation Radiation Monitor (NGRM)** [1-2]:

- **S6-MF/NGRM Level 1 Version** Proton Differential Flux Dataset
- **EDRS-C/NGRM Level 1 Version 3** Proton Differential Flux Dataset

Produced by SPARC [3-4], publicly available through SWE HAPI [5]

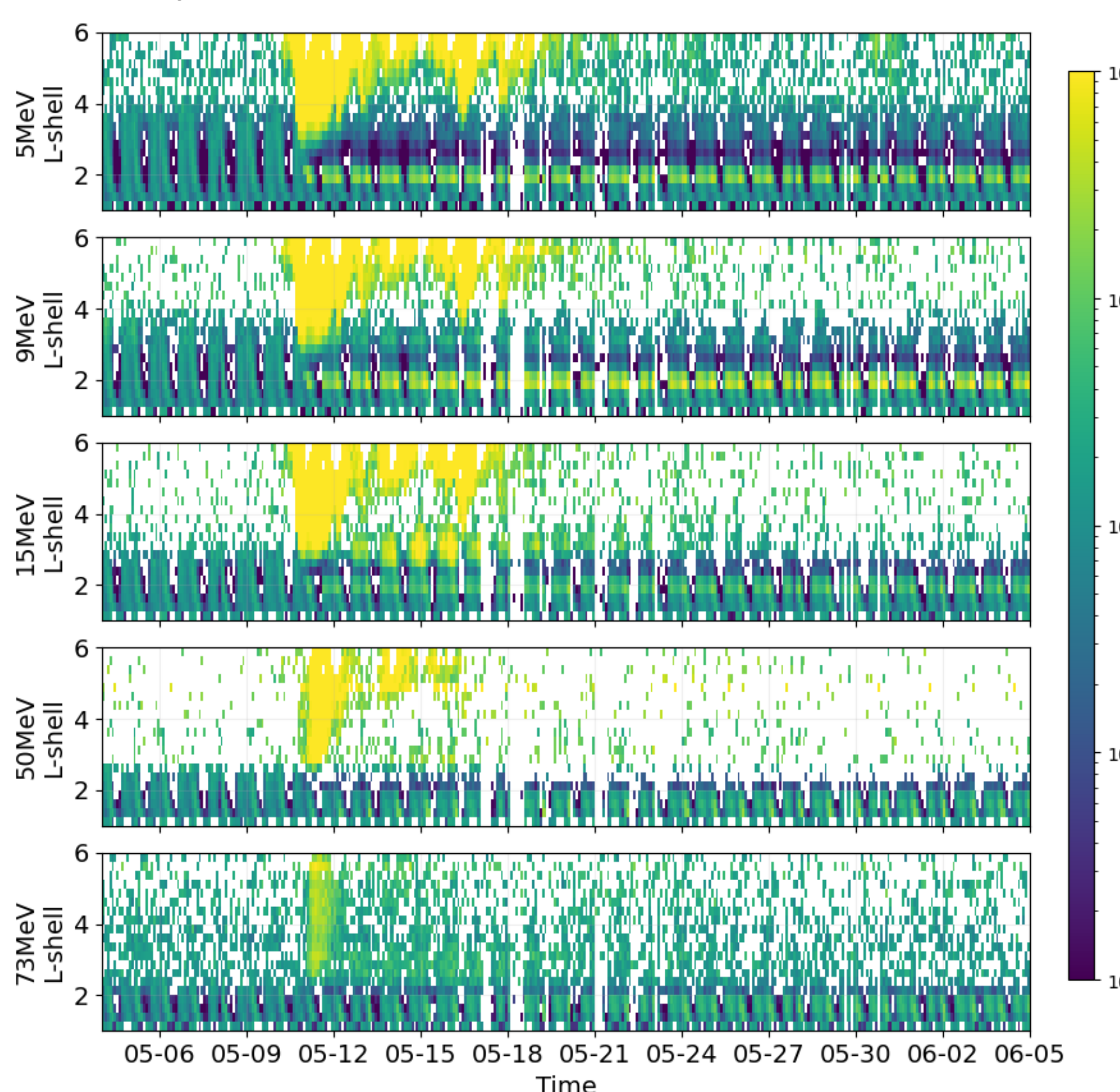


Figure 4: Normalised flux spectrograms for 5 NGRM energy channels, in log scale and in bins 2h long in time and 0.25 in L.

Pre-storm:

- Normal inner belt structure: peak fluxes are around $L \sim 1.25 - 1.5$.
- Higher energy protons are confined at $L = 1.2 - 2.2$.
- Lower energy protons are also visible at higher L-shells (outer part of inner belt and inner part of slot region).
- Counts decrease monotonically with energy at fixed L-shell.

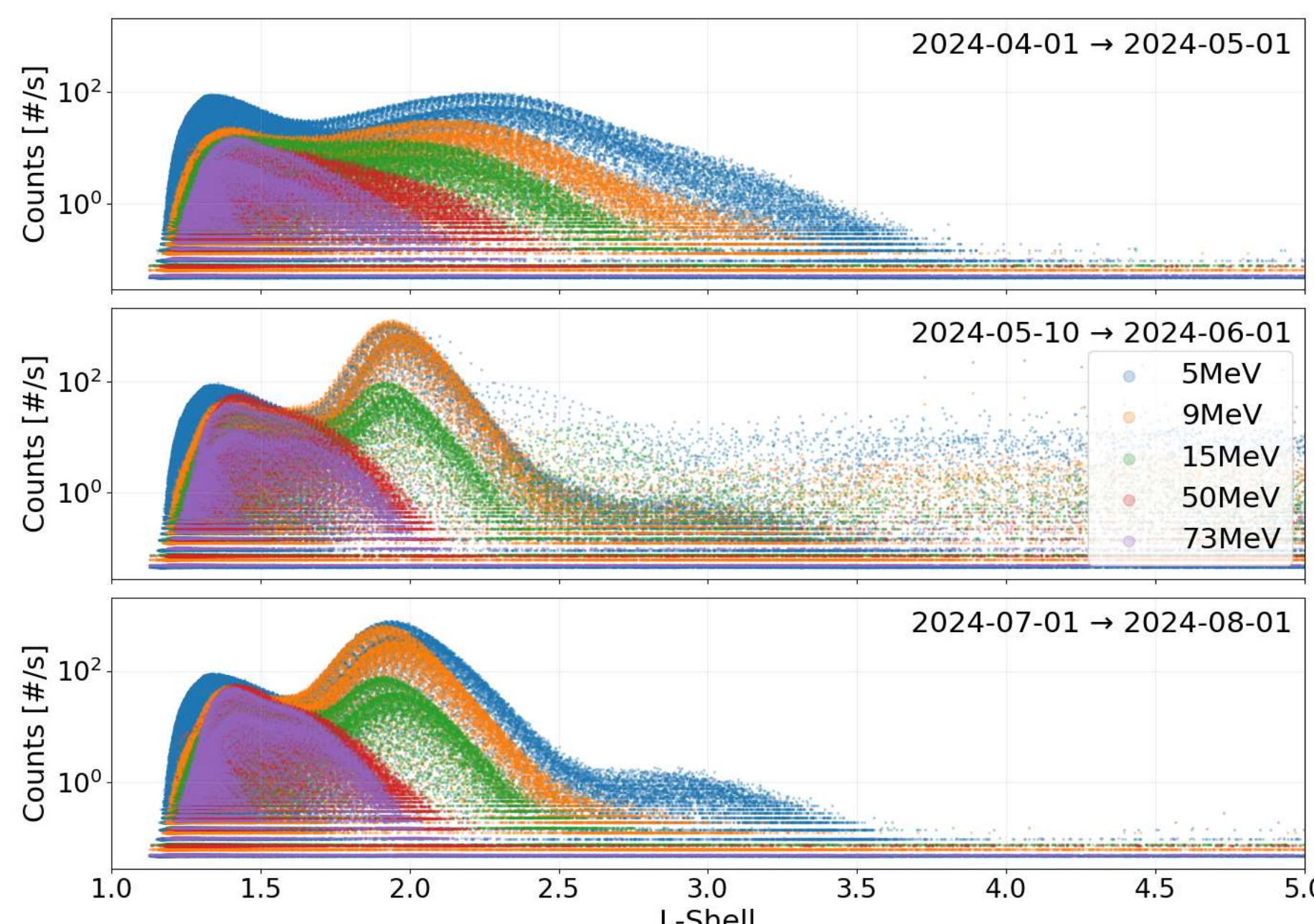


Figure 5: S6-MF/NGRM flux measurements from 5 channels, in log scale, as a function of L, before, during and after the storm.

Storm interval:

- Injected particles are observed by both missions.
- A new - sharply peaked - population of 5, 9 and 15 MeV protons appears at $L \sim 1.9 - 2.1$.
- Higher energy fluxes exceed pre-storm peak levels by roughly an order of magnitude.

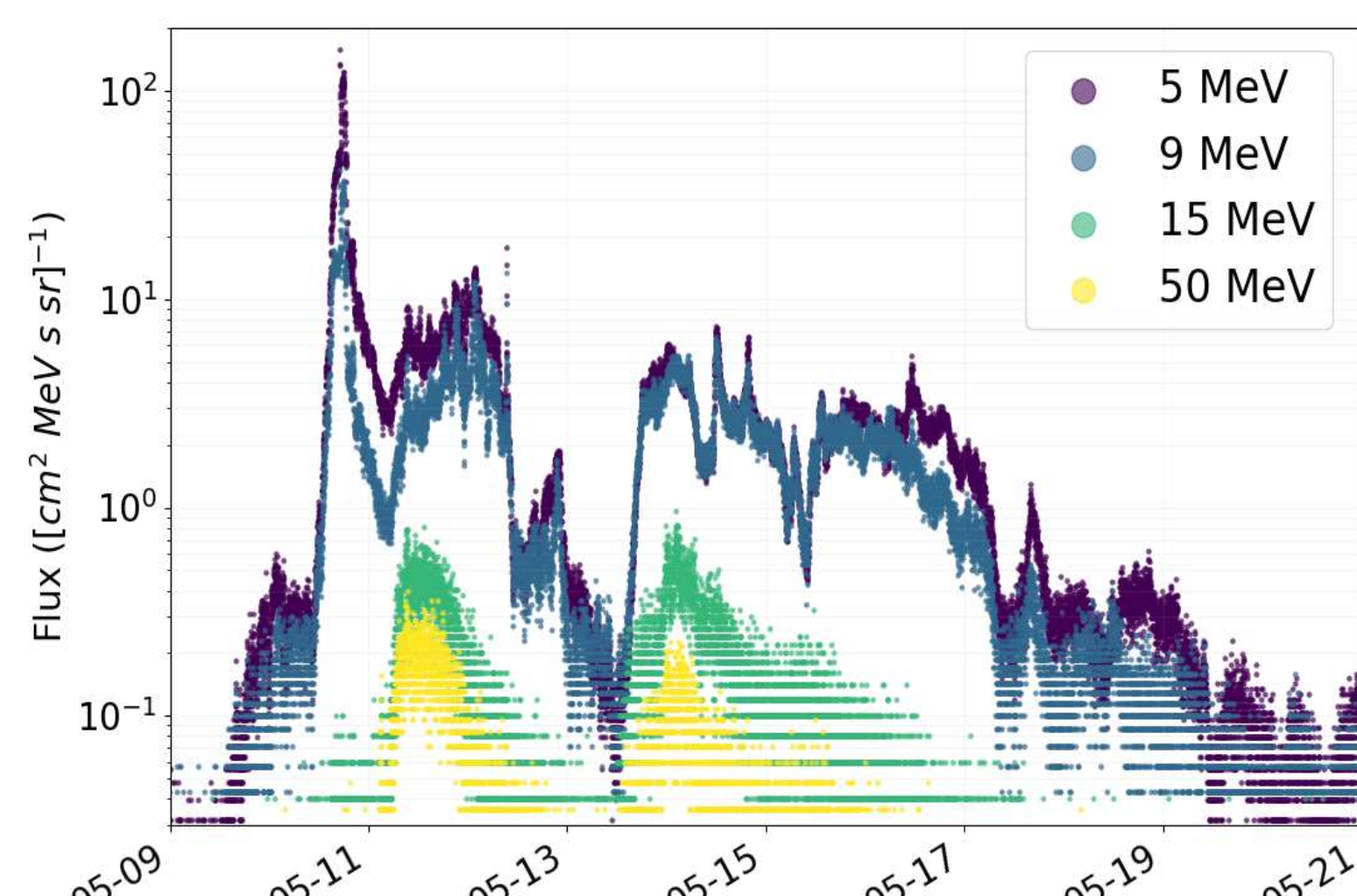


Figure 6: EDRS-C/NGRM flux measurements from 4 channels, in log scale, as a function of time during the storm.

Post-storm:

- The $L \sim 2$ peak persists in the 5, 9, and 15 MeV channels, almost two months after the injection, confirming the long-lived nature of the enhancement.
- Fluxes of 50 and 73 MeV protons at the heart of the belt are still enhanced, consistent with radial diffusion scenario that accelerates the injected particles.
- The double-peaked structure reflects two populations: the stable, but enhanced, inner-belt heart and the newly introduced population at the outer edge of the inner belt and inner edge of the slot region.

Analysis

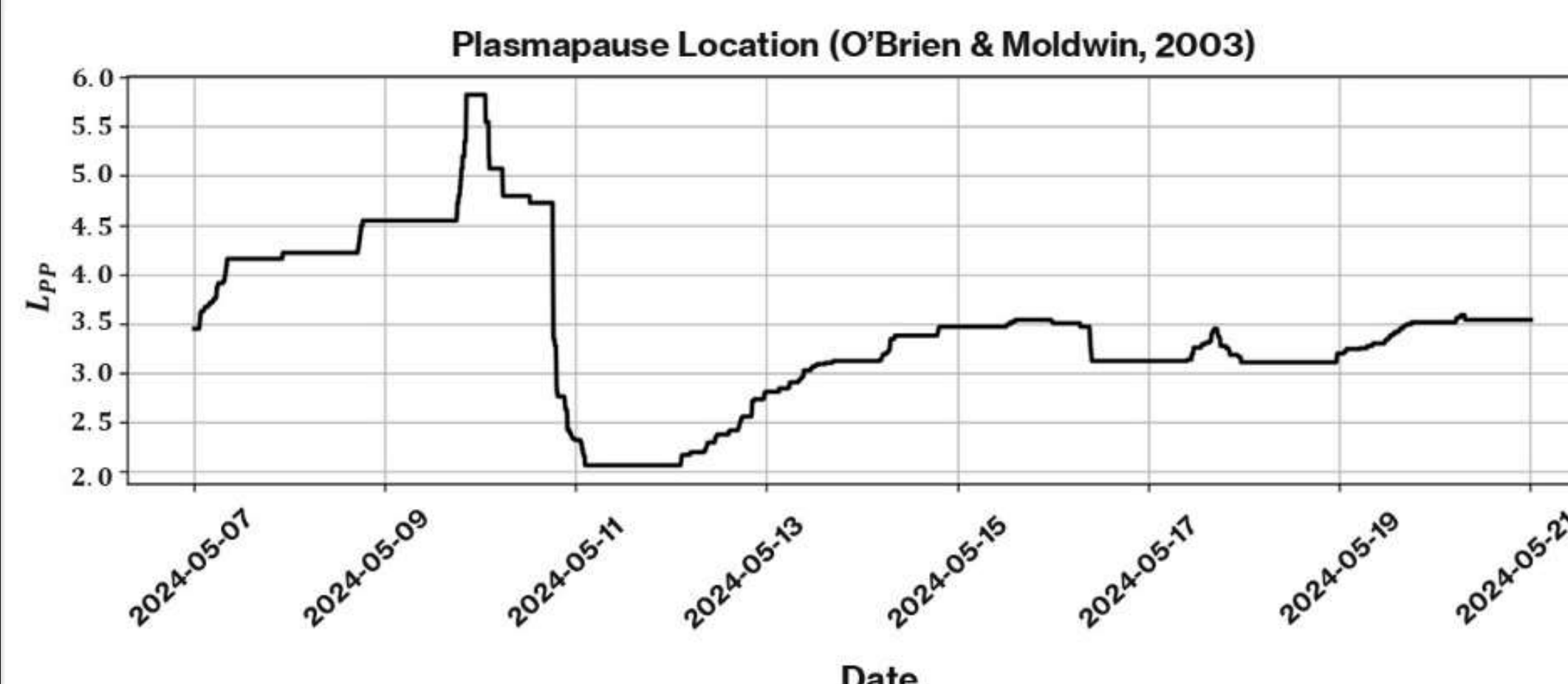


Figure 7: Plasmapause location using the model of O'Brien & Moldwin (2003) [6].

The plasmapause collapses sharply from $L \sim 5.8$ to $L \sim 2$ during storm onset, then gradually recovers to $L \sim 3.5$ over the following week.

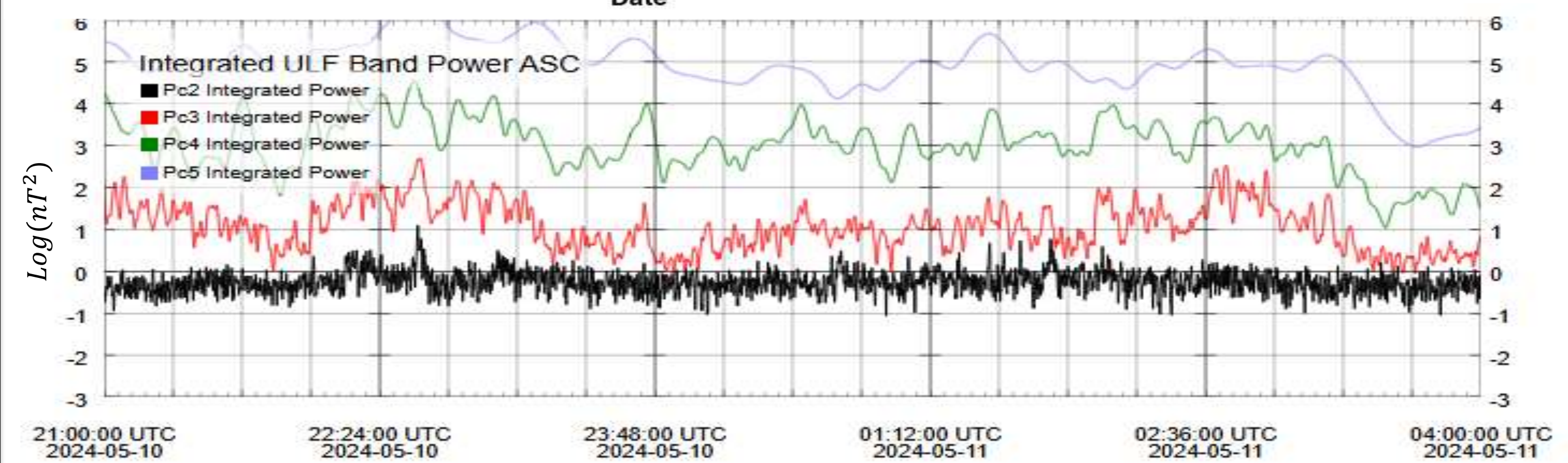


Figure 8: ULF wave activity from SuperMAG data. Bands Pc2-Pc5 shown in different colour.

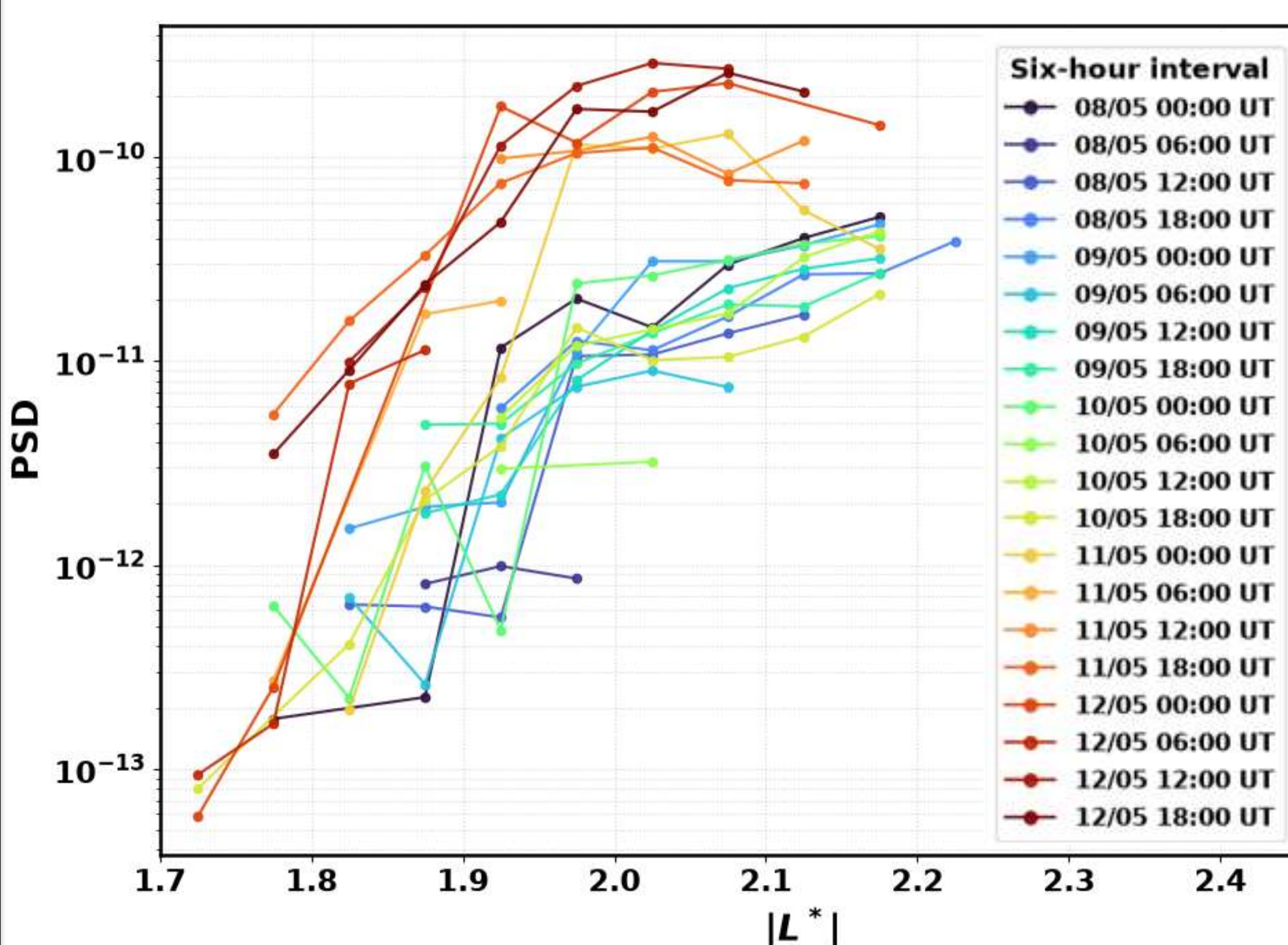


Figure 9: S6-MF 6-hour Radial PSD Profiles with $\mu = 300 \text{ MeV/G}$, $K = 0.375 R_E \sqrt{G}$.

The Pc5 band reached power levels of 10^6 nT^2 , and remained persistently elevated throughout the interval - particularly critical band, because its timescales match the orbital drift periods of MeV protons.

Monotonically positive radial profiles indicate inward radial diffusion of injected particles ranging from $L = 1.8$ and up to $L = 2.1$, consistent with the observed flux enhancement.

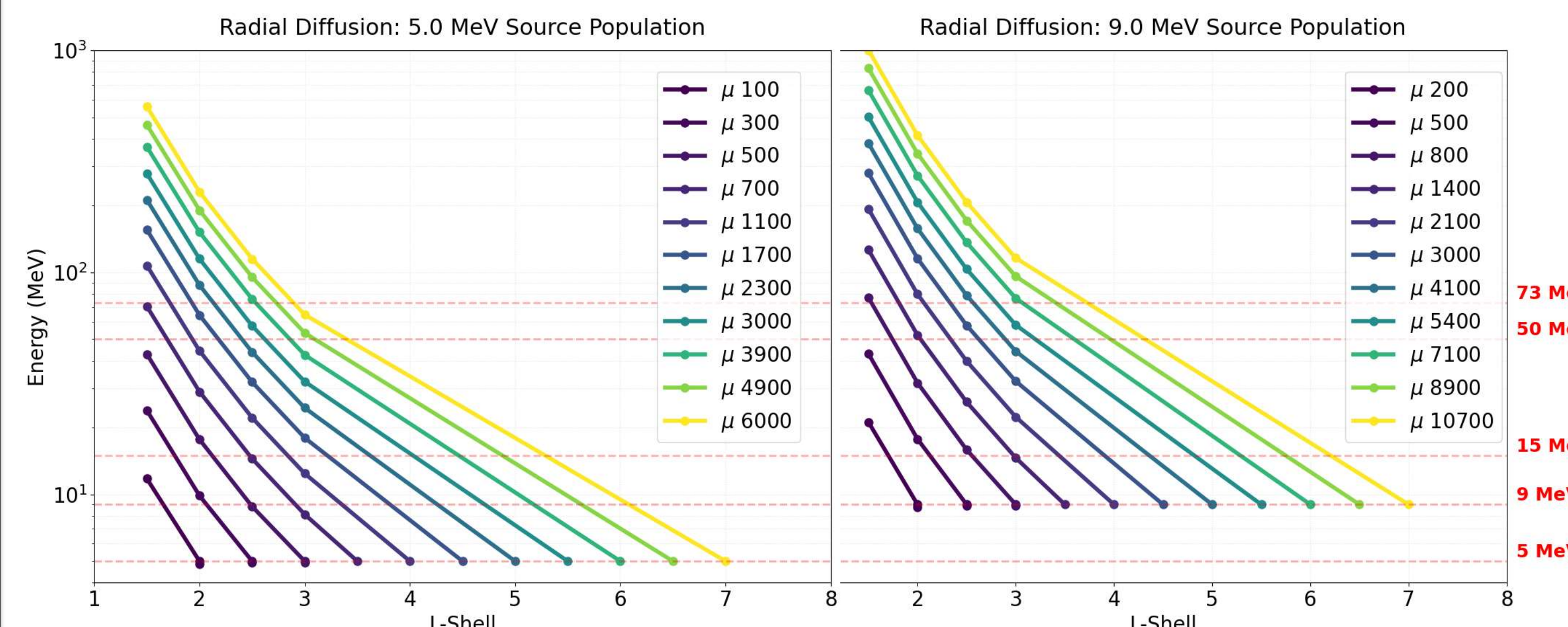


Figure 10: Constant- μ trajectories, for several representative values of the first adiabatic invariant μ , derived from S6-MF/NGRM observations at 5 and 9 MeV, illustrating adiabatic acceleration during inward radial diffusion. The red dashed lines indicate NGRM energy channels.

Simulations

We use a physics-based model [7], that numerically solves the **time-dependent Fokker-Planck equation**, under constant **adiabatic invariants** μ and K , to investigate radial electron transport during the storm.

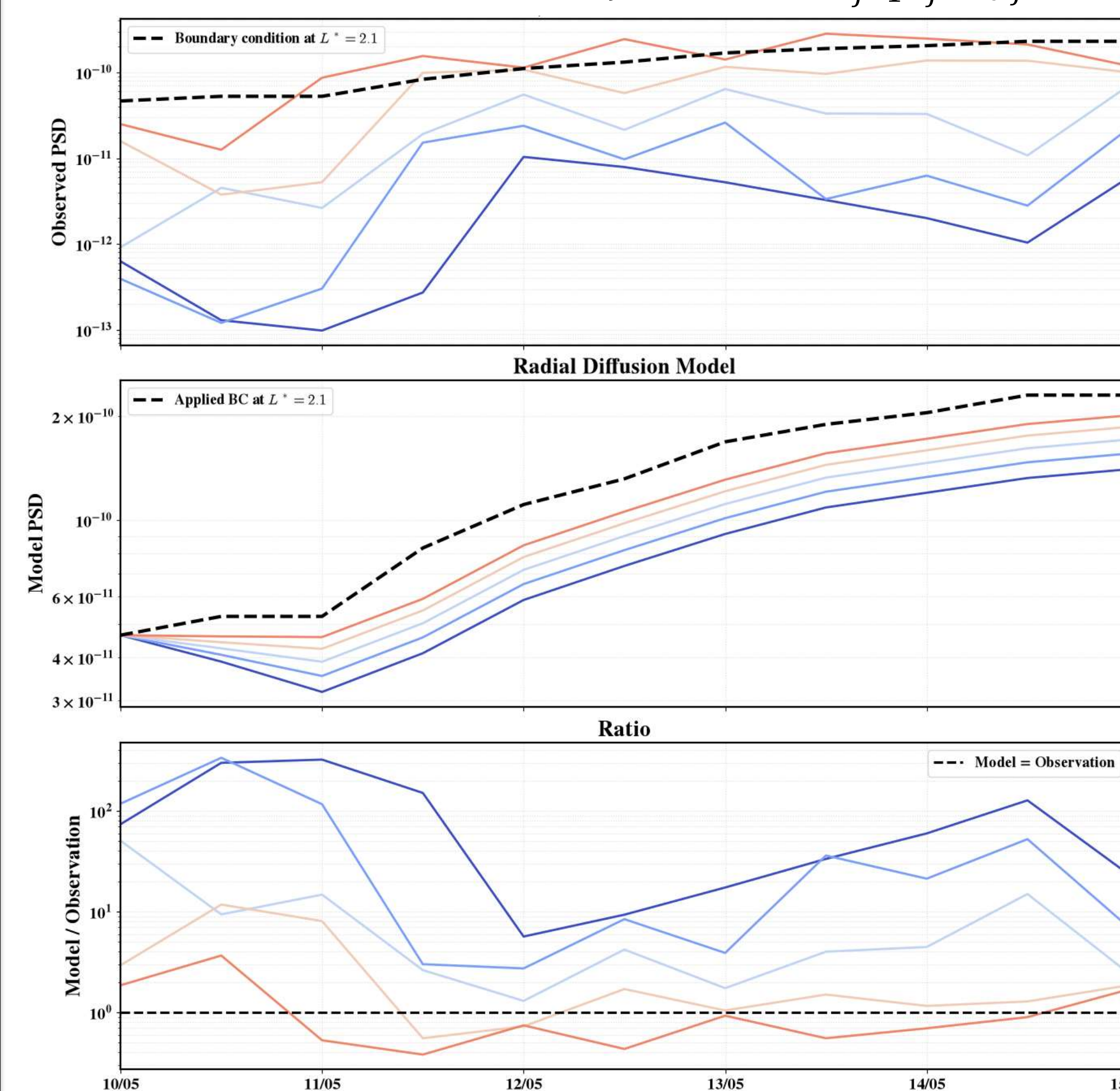
- For the numerical solution, the model uses the **Radial Basis Function (RBF)** collocation methods, enabling comparison of different RBF formulations.

Governing equation: $\frac{\partial f}{\partial t} = L^2 \left(\frac{D_{LL}}{L^2} \cdot \frac{\partial f}{\partial L} \right) - \frac{f}{\tau}$

Data dependency of the model:

Observed PSD sets the outer boundary and the initial condition for all the L^* values.

- ✓ Discretization & Crank-Nicolson time integration
- ✓ Meshless RBF collocation method $f(L, t^{n+1}) \approx \sum_{j=1}^N \lambda_j^{n+1} \varphi_j(r)$



✓ $\tau \rightarrow \infty$ (negligible loss timescale)

✓ D_{LL} from the data driven EMERALD model [8]

✓ Multiquadric RBF $\varphi(r) = \sqrt{r^2 + c^2}$, $r = \|L - L_j\|$; shape parameter $c = 10^{-3}$

✓ Grid: $\Delta t = 12\text{h}$; $\Delta L^* = 0.05$

✓ Adiabatic Invariants: $\mu = 300 \text{ MeV/G}$, $K = 0.375 R_E \sqrt{G}$

Figure 11: Radial Diffusion Model using S6-MF/NGRM Observations for the outer boundary condition.

References:

- [1] Desorgher et al.(2013), "The Next Generation Radiation Monitor—NGRM", IEEE/NSS/MIC.
- [2] Sandberg et al.(2022), "First Results and Analysis From ESA Next Generation Radiation Monitor Unit Onboard EDRS-C", IEEE/TNS, 69(7)
- [3] Sandberg (2023), S6-MF/NGRM Level 1 Data Description, Tech. Rep., ESA.
- [4] Sandberg (2022), EDRS-C/NGRM Level 1 Data Description, Tech. Rep., ESA.
- [5] ESA, Next Generation Radiation Monitor (NGRM), <https://swe.ssa.esa.int/en/ngrm>
- [6] O'Brien and Moldwin (2003), "Empirical plasmapause models from magnetic indices", Geophys. Res. Lett. (30)
- [7] Dimitrakoula et al. (2026), "Physics-based Modelling of the Radial Diffusion in the outer Van Allen belt", in prep.
- [8] Aminiagha-Giamini et al. (2023), "The EMERALD model for the estimation of the radial diffusion coefficients in the outer belt", SW (21)